



Indian Institute of Technology Kharagpur

Class Test II,

Date of Examination:

Duration: 45 Minutes

Subject No.:

Subject:

Department/Center/School: Computer Science & Engg.

Credits:

Full marks: 15

Name: _____

Roll Number: _____

Instructions

- i. This question paper contains 8 pages and 2 questions. All questions are compulsory. Marks are indicated in parentheses. This question paper has been cross checked.
- ii. Please write your name, roll number, subject name and code, date and time of examination on the answer script before attempting any solution.
- iii. Use of electronic calculators only is permitted. No extra resources viz. graph papers, log-tables, trigonometric tables would be required.
- iv. **Organize your work**, in a reasonably neat and coherent way. Work scattered all across the answer script without a clear ordering will receive very little marks.
- v. **Mysterious or unsupported answers will not receive full marks.** A correct answer, unsupported by calculations, explanation, will receive no marks; an incorrect answer supported by substantially correct calculations and explanations may receive partial marks.
- vi. Write your answers in the space provided for that question. If the space provided for the answer is not sufficient, the rest of the answer may be written in the extra space provided after all questions. This should be clearly indicated in the space provided, failing which, the extra answer may not be checked.

Question:	1	2	Total
Points:	7	8	15
Score:			

1. Consider a dataset with four 3-D samples: $\mathbf{x}_1 = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$, $\mathbf{x}_2 = \begin{bmatrix} 1 \\ 4 \\ 2 \end{bmatrix}$, $\mathbf{x}_3 = \begin{bmatrix} 5 \\ 2 \\ 2 \end{bmatrix}$, $\mathbf{x}_4 = \begin{bmatrix} 5 \\ 4 \\ 2 \end{bmatrix}$
- (a) (2 points) Calculate the sample mean $\bar{\mathbf{x}}$ and provide the mean-centered data matrix $\tilde{X}$ where each column corresponds to a centered observation.
 - (b) (2 points) Compute the sample covariance matrix $S = \frac{1}{N-1} \tilde{X} \tilde{X}^T$, where N is the number of samples.
 - (c) (3 points) Find all eigenvalues of S . Identify the eigenvalue and eigenvector associated with the first Principal Component and explain the significance of the last eigenvalue in this specific dataset.

2. A decision tree is being constructed to predict whether a user will subscribe to a service. The target variable is:

$$\text{Subscribe} \in \{\text{Yes}, \text{No}\}$$

At a node of the decision tree, the following subset of data is available for splitting:

Example	Usage Level	Time Spent	Subscribe
D1	High	Long	Yes
D2	High	Long	Yes
D3	High	Short	No
D4	Low	Long	Yes
D5	Low	Short	No
D6	Low	Short	No

While growing the decision tree, two candidate splits are considered at this node.

Split 1: Usage Level

- High $\rightarrow$ (2 Yes, 1 No)
- Low $\rightarrow$ (1 Yes, 2 No)

Split 2: Time Spent

- Long $\rightarrow$ (3 Yes, 0 No)
- Short $\rightarrow$ (0 Yes, 3 No)

(a) (2 points) Compute the entropy of the parent node.

(b) (4 points) Compute the entropy of each child node for both splits.

- (c) (2 points) Based on your results, determine which split the decision tree will choose and justify your answer.

The questions end here.

The next few pages are added extra. You can use these, if for any question you need extra space.
